CAN THO UNIVERSITY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

FACULTY OF SOFTWARE ENGINEERING



GRADUATION THESIS
BACHELOR OF ENGINEERING IN

SOFTWARE ENGINEERING
(HIGH-QUALITY PROGRAM)

RESIDUE-LEVEL DEGRON MUTATION EFFECT PREDICTION AND ANALYSIS TOOL

Student: Nguyễn Thanh Trọng

Student ID: B2305615

Class: 2023 – 2027 (K49)

Advisor: Dr. Ho Quang Thai

Can Tho, [MM/YYYY]

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CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

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THESIS REVISION CONFIRMATION IN ACCORDANCE WITH THE COMMITTEE'S COMMENTS

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Student: Nguyễn Thanh Trọng Student ID: B2305615

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Supervisor

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COMMENTS OF THE ADVISOR

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Advisor

Dr. Ho Quang Thai

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Student

Nguyễn Thanh Trọng

COMMITMENT TO RESULTS

I hereby certify that the graduation thesis titled "RESIDUE-LEVEL DEGRON MUTATION EFFECT PREDICTION AND ANALYSIS TOOL" is the result of my own research and has not been submitted for any other degree or qualification.

All references used in the thesis are clearly cited, including the author's name, the title of the work, and the time and place of publication. I accept full responsibility for any unauthorized copying, violations of academic regulations, or instances of academic dishonesty.

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ABSTRACT

A degron is a sequence element that marks a protein for selective degradation, and a single amino acid substitution within it can change its activity. Experiments score whole peptides, while measuring every substitution is costly. This thesis studies how to predict the effect of single amino acid substitutions in degron peptides at residue level from experimental stability data, evaluates generalisation with protein-cluster-aware splits against ridge, PAP-difference and Degformer baselines, and builds a prototype tool that displays predicted effect matrices, ranks substitutions and exports results with provenance. **[PENDING: main results and limitations, to be written when the work is complete.]**

Keywords: degron, protein stability, mutation effect prediction, ubiquitin–proteasome system, machine learning.

TÓM TẮT

Degron là đoạn trình tự đánh dấu protein để bị phân hủy có chọn lọc; một phép thay thế amino acid trong degron có thể làm thay đổi hoạt tính này. Thực nghiệm thường chấm điểm cả peptide, trong khi đo mọi phép thay thế rất tốn kém. Luận văn nghiên cứu cách dự đoán hiệu ứng của phép thay thế đơn amino acid trong peptide degron ở mức residue từ dữ liệu độ bền protein thực nghiệm, đánh giá khả năng tổng quát hóa khi chia dữ liệu theo cụm protein so với các đối chứng ridge, chênh lệch điểm PAP và Degformer, đồng thời xây dựng nguyên mẫu hiển thị ma trận hiệu ứng, xếp hạng phép thay thế và xuất kết quả kèm nguồn gốc. **[CHỜ BỔ SUNG: kết quả chính và giới hạn, viết khi hoàn thành công việc.]**

Từ khóa: degron, độ bền protein, dự đoán hiệu ứng đột biến, hệ ubiquitin–proteasome, học máy.

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GLOSSARY AND ABBREVIATIONS

No.	Term / Abbreviation	Definition / Explanation
1	Degron	Sequence element that marks a protein for selective degradation.
2	E3 ligase	Enzyme that recognises substrate proteins and transfers ubiquitin to them.
3	Residue	One amino acid at a specific position of a protein or peptide.
4	Footprint	Map of positions or substitutions that affect degron activity under a given assay.
5	GPS	Global Protein Stability profiling, a fluorescent-reporter assay of protein stability.
6	PSI	Protein Stability Index, a summary of the reporter signal distribution over sorting bins.
7	ΔPSI	PSI of the mutant minus PSI of the wild type.
8	WT	Wild type, the unmutated reference sequence.
9	PAP	Model that predicts degron activity of a whole peptide.
10	AUROC	Area under the receiver operating characteristic curve.
11	MMseqs2	Software for sequence search and clustering.
12	Cov80 / cov50	Protein clusters built with 80% / 50% alignment coverage.

PART 1. INTRODUCTION

1. Problem Statement

The abundance of a protein in a cell depends on both its synthesis and its degradation. In the ubiquitin–proteasome system, a *degron* is a short sequence element that marks a protein for selective degradation [1], [2]. A single amino acid substitution inside a degron can strengthen or weaken this signal, so knowing the effect of each substitution helps researchers choose mutations to test experimentally [3], [4].

Existing experiments score a whole peptide with one activity value. That value does not say which position, or which substituted amino acid, makes the degron signal stronger or weaker. Measuring every substitution experimentally is costly, which motivates a predictive model that shortlists substitutions. Such a model is only useful if it is evaluated on experimental labels and if similarity between training and test proteins is controlled.

2. History of Approaches to the Problem

2.1. International Context

Large-scale stability profiling based on the Global Protein Stability (GPS) assay [5] has produced experimental degron maps, including scanning and saturation mutagenesis of human peptides [3], cytosolic tile maps [4] and transcription-factor tile maps [6]. On the computational side, PAP and Degformer predict peptide-level degron activity or protein stability [4], [7], MetaDegron predicts E3-ligase-targeted degron motifs [8], and DEGRONOPEDIA and DegronMD provide annotation resources [9], [10].

These works address peptide- or motif-level prediction and annotation. Residue-level substitution effects measured on a common assay endpoint, evaluated with protein-cluster-aware splits, remain a question that the thesis examines rather than a gap it assumes is unfilled.

2.2. Domestic Context

[PENDING: literature search on degron and protein-stability prediction work in Vietnam; to be written only after the search is done.]

3. Objectives

3.1. General Objective

To study methods that predict the effect of amino acid substitutions in degrons from experimental data, to evaluate generalisation across protein clusters, and to build a tool that supports analysis and prioritisation of mutations.

3.2. Specific Objectives

1. Build a reproducible pipeline that normalises data, maps coordinates and derives mutation-effect labels while keeping each source and assay condition separate.
2. Establish baselines: amino acid composition, ridge regression, the PAP score difference and the Degformer architecture, under clearly stated comparable conditions.
3. Test whether representations learned from peptide-level labels and fine-tuned on mutation labels improve prediction, and study the learning curve before choosing a model.
4. Evaluate effectiveness, stability across seeds, error, and the contribution of each component.
5. Build a software prototype that accepts a peptide, shows the predicted effect matrix, ranks substitutions and exports results with model information.

4. Research Subject and Scope

4.1. Research Subject

Degron peptides, single amino acid substitutions, protein stability indices measured with a reporter assay, and models predicting residue-level mutation effects. A *residue* is one amino acid at a specific position; the *footprint* is the map of positions or substitutions that change degron activity under a given assay.

4.2. Research Scope

- The main label source is the human peptide data of Zhang et al. [3]; other peptide-level sources are used only for transfer learning or domain studies when

their terms of use allow.

- Input: reference (wild-type) sequence, position and substituted amino acid. Output: predicted effect ΔPSI, which can be arranged into an $L \times 20$ matrix.
- Identifying the specific E3 ligase is not a main task. No new wet-lab experiments are performed, and software output supports hypothesis generation; it does not establish causation, disease effects or therapeutic benefit.

5. Research Process and Technologies Used

5.1. Research Process

The work combines a literature review, controlled computational experiments and a software prototype: (1) study degron biology and related methods; (2) audit data and design the evaluation; (3) train and compare models; (4) develop and test the prototype; (5) analyse results and write the thesis.

5.2. Technologies Used

Python with NumPy, pandas, scikit-learn and PyTorch for modelling, and MMseqs2 for protein clustering [11]. **[PENDING: prototype stack, to be fixed when the system design is chosen.]**

5.3. Development and Supporting Tools

Git for version control, LaTeX (XeLaTeX) for this report. **[PENDING: testing and deployment tools.]**

6. Main Contributions

[PENDING: to be written from achieved results at the end of the work. Planned contributions are a reproducible data and evaluation pipeline, a documented comparison with baselines including negative results, and a prototype tool.]

7. Structure of the Thesis

Part 1 introduces the problem, objectives and scope. Part 2 has four chapters: Chapter 1 describes the problem and the system requirements; Chapter 2 gives the

theoretical background; Chapter 3 presents the design of the data, method and system; Chapter 4 covers experimental evaluation and software testing. Part 3 states the results, limitations and future work. The appendices give installation and usage instructions and supplementary material.

PART 2. CONTENT

CHAPTER 1: PROBLEM DESCRIPTION

1.1. Detailed Problem Description

Researchers who study degrons want to know which substitution, at which position of a degron peptide, raises or lowers the degron signal. The thesis addresses two connected problems.

- **Prediction.** Given a reference peptide sequence, a position and a substituted amino acid, predict the effect ΔPSI of the substitution. Models are trained and evaluated on experimental labels with protein-cluster-aware splits.
- **Analysis tool.** Provide a prototype that lets a user enter a peptide, view the predicted effect matrix, rank substitutions and export the result with provenance (model version, data source, supported domain).

The intended users are researchers who select mutations for follow-up experiments, and students who explore the relation between peptide sequence and degradation signal. The workflow is: enter a valid peptide, generate all substitutions, score them with a fixed model, display a matrix and ranking, and export CSV/JSON. Inputs that are missing, mis-positioned or outside the supported domain must be reported explicitly, and uncalibrated scores must not be presented as probabilities of disease.

1.2. Functions of the System

A minimal offline command-line prototype exists (Section 4.3); it covers scoring and CSV export only, with no web interface yet. The use cases below remain the requirements for the full prototype.

1.2.1. Predict the Effect Matrix

Table 1.1. Predict the effect matrix of a peptide

Use case name: Predict effect matrix	**ID:** UC-01
Primary actor: User	**Necessity level:** Important **Classification:** Simple

Stakeholders and interests: Researcher: wants predicted effect of every substitution of a peptide.
Summary description: The user enters a peptide sequence; the system returns an $L \times 20$ matrix of predicted ΔPSI values.
Relationships: Association: User Include: Validate input Extend: Compare with measured data Generalization: —
Event handling: 1. The user enters a peptide. 2. The system validates length, alphabet and domain. 3. The system scores all substitutions with the fixed model. 4. The system displays the matrix.
Alternative / exceptional flows: Invalid or out-of-domain input: the system shows an explicit message and does not return scores.
Expected outcome: A matrix of predicted effects with model version and supported domain.

1.2.2. Rank Substitutions

Table 1.2. Rank substitutions

Use case name: Rank substitutions	**ID:** UC-02
Primary actor: User	**Necessity level:** Important **Classification:** Simple
Stakeholders and interests: Researcher: wants a short list of substitutions to test.	
Summary description: The system orders substitutions by predicted effect.	
Relationships: Association: User Include: Predict effect matrix Extend: — Generalization: —	

Event handling: 1. The user opens the ranking of a predicted peptide. 2. The system sorts substitutions by predicted effect. 3. The user filters by position or amino acid.
Alternative / exceptional flows: No prediction available: the system asks the user to run UC-01 first.
Expected outcome: A ranked table of substitutions.

1.2.3. Export Results with Provenance

Table 1.3. Export results with provenance

Use case name: Export results	**ID:** UC-03
Primary actor: User	**Necessity level:** Medium **Classification:** Simple
Stakeholders and interests: Researcher: wants results that can be reused and traced.	
Summary description: The system exports matrix and ranking with model version, data source and supported domain.	
Relationships: Association: User Include: Predict effect matrix Extend: — Generalization: —	
Event handling: 1. The user selects an export format (CSV or JSON). 2. The system writes the file with provenance fields.	
Alternative / exceptional flows: Export fails: the system reports the error and keeps the result on screen.	
Expected outcome: A file containing scores and provenance.	

1.3. Non-Functional Requirements

These are targets for the prototype; **[PENDING: concrete thresholds to be fixed with the system design.]**

1.3.1. Performance Requirements

Scoring a single peptide should finish in interactive time on a standard laptop (target to be measured).

1.3.2. Information Safety Requirements

Only public data are used; terms of use of each source are recorded before redistribution.

1.3.3. Security Requirements

Input is validated; user-supplied sequences are not executed or stored beyond the session unless stated.

1.3.4. Reliability Requirements

The same input and model version must give the same scores; scores shown in the interface must match the research runner within numerical tolerance.

1.3.5. Maintainability Requirements

Models and data are versioned; experiments can be re-run from documented commands.

1.4. Implementation and Design Constraints

1.4.1. Implementation Constraints

The work is done by one student within the thesis course schedule; no new wet-lab experiment is performed; computation runs on CPU-class hardware.

1.4.2. Design Constraints

Scores are reported for the assay and domain on which the model was evaluated; no claim is made on disease causation or treatment.

CHAPTER 2: THEORETICAL BACKGROUND

2.1. Domain Background

2.1.1. Proteins, Amino Acids and Residues

A protein is a chain of amino acids. A *residue* is one amino acid at a specific position of the chain. A *single amino acid substitution* replaces the residue at one position with another of the 20 standard amino acids.

2.1.2. The Ubiquitin–Proteasome System and Degrons

In the ubiquitin–proteasome system, E3 ubiquitin ligases recognise substrate proteins and attach ubiquitin chains to them; ubiquitinated proteins are then degraded by the proteasome. A *degron* is a sequence element that is part of this recognition and marks a protein for selective degradation [1], [2]. Degrons can lie at the protein termini or inside the sequence [3]. The activity of a peptide taken out of its protein may differ from its activity in the intact protein, because surface exposure, structure and cellular context matter.

2.1.3. Measuring Degron Activity

Global Protein Stability (GPS) profiling [5] fuses a peptide to a fluorescent reporter and sorts cells by fluorescence into bins. The *Protein Stability Index* (PSI) summarises the distribution of cells over the bins; in the data of Zhang et al. [3], a lower PSI corresponds to a less stable reporter. The effect of a substitution is defined as

$$\Delta\mathrm{PSI} = \mathrm{PSI}_{\mathrm{mutant}} - \mathrm{PSI}_{\mathrm{WT}}, \tag{2.1}$$

where a positive value means the reporter is more stable under the assay. This describes the measured reporter stability and does not by itself show direct recognition by a specific E3 ligase.

Scanning mutagenesis replaces residues in a systematic way along the peptide, whereas *saturation* mutagenesis tests many amino acids at each position. Cells from the same peptide are statistically related and cannot be treated as fully independent samples. Measurement noise comes from read counts, replicate variation and experimental conditions.

2.2. Methods and Models

2.2.1. Problem Formulation

The task is to predict ΔPSI for a (peptide, position, substituted amino acid) triple. Predictions for all positions and amino acids form a matrix of size $L \times 20$ for a peptide of length L. For evaluation, ΔPSI can also be thresholded into a binary label; the threshold and the direction of the effect must be fixed in the evaluation protocol before models are compared.

2.2.2. Baseline and Reference Models

- **Ridge regression** [12]: a linear model with L_2 regularisation, used as a simple baseline on encodings of the reference and substituted amino acids, peptide composition and local context.
- **PAP score difference**: PAP predicts a degron score for a whole peptide [4]; the mutation effect is taken as $\mathrm{PAP}_{\mathrm{mutant}} - \mathrm{PAP}_{\mathrm{WT}}$.
- **Degformer**: a transformer-based model [13] that predicts PSI [7].

MetaDegron [8] predicts degron motifs and is used to compare problem scope, not as a label source.

2.2.3. Transfer Learning

Transfer learning pre-trains a representation on peptide-level activity labels and then fine-tunes it on mutation-level labels. The two label types have different meanings, so they need separate prediction heads or a suitable transformation; binary motif labels are not converted into ΔPSI values.

2.3. Technologies and Tools

Python with NumPy, pandas and scikit-learn for data handling and classical models; PyTorch for neural models; MMseqs2 [11] for clustering protein sequences. Versions are recorded when experiments are run. **[PENDING: web/API stack for the prototype.]**

2.4. Evaluation Methodology

2.4.1. Splitting by Protein Cluster

Randomly splitting individual cells lets similar sequences appear in both training and test sets and inflates results. Proteins are therefore clustered by sequence similarity (MMseqs2, 30% identity, 80% coverage as the main setting; 50% coverage as a sensitivity check) and whole clusters are assigned to folds. Hyperparameters are chosen on a cluster-wise validation set inside each training fold; scaling, feature selection and pre-training must not use the held-out clusters.

2.4.2. Metrics

The main metric is the cell-level area under the ROC curve (AUROC), computed per peptide that contains both classes and then averaged. Spearman correlation or mean absolute error on continuous ΔPSI is added when labels allow.

2.4.3. Statistical Comparison

Models are compared pairwise on the same peptides. Confidence intervals come from bootstrap resampling [14] by protein cluster, and results are reported per random seed in addition to the aggregate. A development benchmark is distinguished from an independent validation.

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

This chapter gives the planned structure of the solution. Details of implementation are **[PENDING]** until the corresponding work is done.

3.1. System Overview

3.1.1. Product Context

The system has two parts: a research pipeline (data preparation, training, evaluation) and a prototype tool that uses a fixed trained model to score substitutions.

3.1.2. Functional Overview

Functions follow the use cases of Chapter 1: predict an effect matrix, rank substitutions, export results with provenance.

3.2. System Architecture

3.2.1. Architectural Components

[PENDING: architecture diagram and component roles.]

3.2.2. Interaction Between Components

[PENDING]

3.2.3. Design Rationale

[PENDING: advantages and drawbacks of the chosen architecture.]

3.3. Data Design

3.3.1. Data Sources and Preprocessing

Table 3.1 lists the planned sources. Each source is pinned by version or download date, checksum and terms of use; labels from different assays are kept separate.

3.3.2. Data Model

[PENDING]

Table 3.1. Planned data sources

Source	Role
Zhang et al. 2023, Mendeley Data v2 [15]	Main labels: scanning and saturation mutagenesis
GEO GSE240610 [16]	Cross-check of the peptide library source
Voutsinos et al. [4]	Peptide-level tiles and PAP, for baselines and transfer learning
Larsen et al. [6]	Domain-shift study at tile level, if terms of use allow
UniProtKB/Swiss-Prot	Sequences and coordinates for mapping and clustering

3.3.3. Data Dictionary

[PENDING: table of fields.]

3.3.4. Relationships Between Entities

[PENDING]

3.4. Method and Model Design

Input is the reference sequence, position and substituted amino acid; target is ΔPSI. The baseline is ridge regression on amino acid encodings, composition and local context; the reference models are the PAP score difference and Degformer. Transfer learning from peptide-level labels is examined against random initialisation under the same folds and label budget. **[PENDING: final architecture, objective function and training configuration.]**

3.5. Functional Design

[PENDING: processing flow and interface for each function.]

9

CHAPTER 4: TESTING AND EVALUATION

4.1. Testing Objectives

To check that the research pipeline gives reproducible results without leakage between training and test clusters, and that the prototype validates input, keeps coordinates and score direction correct, and exports data consistently with the research runner.

This chapter has two parts. Section 4.2 reports the experimental evaluation of the research pipeline; these experiments have been run. Sections 4.3 to 4.6 follow the test-plan outline of the Faculty for the prototype tool. Only a minimal offline command-line prototype exists (no graphical interface); its consistency checks against the research runner have been run and are reported there, while the parts of the plan that need the full prototype are still **[PENDING]**.

4.2. Experimental Evaluation

4.2.1. Experimental Setup

Data and labels. The labels come from the saturation-mutagenesis experiment of Zhang et al. [3], [15]. For each peptide, the effect of a substitution is $\Delta\text{PSI} = \text{PSI}_{\text{mutant}} - \text{PSI}_{\text{WT}}$ (Chapter 2). The data set contains two replicate halves of each peptide. The PSI_{WT} reference of a half is the median PSI of the synonymous cells of that half, and at least ten such cells are required. The cell label is the mean ΔPSI of the two halves; a cell is *positive* when $\Delta\text{PSI} \geq 1$. A peptide is accepted as a degron only if $\text{PSI}_{\text{WT}} \leq 3.0$. Table 4.1 shows how the 248 raw saturation peptides reduce to the 91 peptides of the benchmark. The single largest cause of loss is the degron threshold, which removes 144 peptides; the technical causes together remove only $10 + 2 + 1 = 13$ peptides. The benchmark set contains 91 peptides in 89 protein clusters and 47 847 cells. The measurement ceiling of this set, computed as the agreement between its two experimental halves, is 0.9519 AUROC.

Splits. Proteins are clustered with MMseqs2 [11] at 30% identity and 80% coverage (cov80). The outer evaluation uses five folds with whole clusters assigned to folds. Inside each outer training set, 20% of the clusters form a validation set for choosing

Table 4.1. Fate of the raw saturation peptides under the label filters

Outcome of the label filters	Peptides
Retained: the benchmark set	91
Removed only by the degron threshold ($PSI_{WT} > 3.0$)	144
Fewer than 10 synonymous cells for the WT reference in a half	10
Sequence mismatch between scan file and UniProt	2
No sequence could be derived	1
Raw saturation peptides	*248*

Source: `next-steps-plan-2026-10-22/sat-filter-dropoff.json`.

hyper-parameters, so no outer-test label is ever used for tuning. Training is repeated with seeds 17, 29 and 43. The three seeds share the same outer partition and differ only in the inner validation split and in network initialisation, so they are not three independent partitions. The reported score of a model is the AUROC of the average of the three seeds' out-of-fold scores; per-seed results are also kept.

Models.

- **Ridge M3.** Ridge regression [12] on 180 features: one-hot reference and substituted amino acid (40), peptide composition (20) and the ordered ± 3 residue context (120). The penalty is chosen from $\{1, 10, 100, 1000\}$ on the inner validation set.
- **ΔPAP.** The difference of the PAP degron score [4] between mutant and wild type, with the published network frozen (variant `cnn2w1`). It is the reference baseline and uses no training labels.
- **Gated networks.** The same 180 features, with the substitution (pair) and the composition-plus-context inputs projected separately to 32 dimensions; a sigmoid of the pair projection multiplies the context representation, followed by an MLP ($64 \to 32 \to 1$). Variants add the frozen ΔPAP as a residual or remove the gate. Training uses AdamW (learning rate 2×10^{-3}, weight decay 0.01, batch of about 1,024, at most 60 epochs, patience 8).
- **Degformer.** The published checkpoint (epoch 50) of the transformer [7], scored as $\text{pred}(\text{mutant}) - \text{pred}(\text{WT})$; and the same architecture (d_model 128, 4 layers,

8 heads) retrained from scratch on the training clusters of each fold.

Further interaction designs (FiLM, rank-8 bilinear, ridge with cross-product features) were added as *pre-registered candidates* in Section 4.2.3.3.

Pre-registration. The acceptance criteria of the overlap check, the interaction designs, the retrained Degformer architecture, the learning-curve and threshold experiments, the re-clustering, the pre-training experiment, the calibration and the audit of the external labels were written to a protocol file before the experiments were run, and were not changed afterwards. Where a result below is described as met or not met, it refers to those fixed criteria. Observations made after seeing the numbers are labelled as descriptive. The first direct-learning experiment (Table 4.2) has a protocol file but is treated as exploratory.

Environment. All experiments ran on CPU with Python 3.12.3, NumPy 2.5.3 and PyTorch 2.14.0 (CPU build); each result file records the SHA-256 of its input and of its runner. Large caches and checkpoints are generated again by the runners and are not kept in the repository.

4.2.2. Evaluation Metrics

The main metric is the cell-level AUROC (positive: $\Delta\text{PSI} \geq 1$), computed per peptide that has both classes and then averaged over peptides (macro AUROC). Paired differences between two models are computed on the same peptides. The 95% confidence intervals come from 5,000 bootstrap resamples of protein clusters [14]; every peptide of a sampled cluster is kept, so cells and peptides are not treated as independent. These intervals are conditional on the stored out-of-fold predictions and do not include the variance of retraining. No correction is made for the many comparisons, so a single interval is a development-stage indication and not a confirmation.

Probabilities are assessed with the Brier score (the mean squared difference between the predicted probability and the 0/1 label over the cells), the Brier skill score against a constant forecast of the base rate, $\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{const}}$, and the expected calibration error (ECE) over ten equal-mass bins, pooled over the outer-test cells and bootstrapped by cluster in the same way. Ranking utility is the *enrichment at* $k\%$: for each peptide, the substitutions are ordered by predicted score, the precision

of the top $k\%$ is divided by the fraction of positive cells of that peptide, and the ratio is averaged over the peptides that have both classes; a random ranking has enrichment 1.

4.2.3. Results

All results below are on a *development benchmark*: the same 91 peptides have been examined in several experiments, and the out-of-fold design is not an independent validation on unseen data.

4.2.3.1. Direct Learning on the Saturation Matrix

Table 4.2 lists the macro AUROC of each model. Ridge M3 reaches 0.8082, above the frozen ΔPAP baseline (0.7693). The paired difference is +0.0389 with a cluster-bootstrap interval [+0.0216, +0.0571] (Table 4.3). The gated network with residual ΔPAP reaches 0.8002 and is below ridge (difference −0.0080, [−0.0152, −0.0006]). Removing the gate (0.7937) or the ΔPAP residual (0.8015) does not change this picture: both variants stay below ridge.

Table 4.2. Macro AUROC on the 91 peptides of the saturation benchmark. The last column keeps only the cells whose exact sequence is not in the training data of the published Degformer.

Model	All cells	Cells without exact-sequence overlap
ΔPAP, frozen (reference)	0.7693	0.7531
Ridge regression on M3 features	0.8082	0.8040
Gated network + residual ΔPAP	0.8002	0.7956
Network without gate, with ΔPAP	0.7937	0.7892
Gated network without ΔPAP	0.8015	0.7940
Degformer, published checkpoint (overlaps its training data)	0.9248	0.9227

Source: `g2-saturation-2026-10-18/results.json`, `g2-extension-2026-10-18/e1-overlap.json`.

The result of the gate is not stable across seeds (Table 4.4). In seed 29 the gated network is *below* the network without the gate (difference −0.0076, [−0.0152, −0.0005]), whereas it is above in the other two seeds. The ensemble figure should therefore not be read as evidence that the gate helps.

Table 4.3. Paired AUROC differences between models

Paired comparison (same peptides)	Difference in AUROC [95% cluster-bootstrap CI]
Ridge − ΔPAP	+0.0389 [+0.0216, +0.0571]
Gated + ΔPAP − ΔPAP	+0.0309 [+0.0125, +0.0501]
Gated + ΔPAP − ridge	−0.0080 [−0.0152, −0.0006]
Gated + ΔPAP − no gate	+0.0065 [+0.0027, +0.0102]
Gated + ΔPAP − gated without ΔPAP	−0.0013 [−0.0048, +0.0023]

Source: `g2-saturation-2026-10-18/audit.json`, `g2-saturation-2026-10-18/results.json`.

Table 4.4. Per-seed AUROC of the three main models

Seed	Ridge	Gated +ΔPAP	No gate	Gated − no gate [95% CI]
17	0.8062	0.7895	0.7806	+0.0089 [+0.0038, +0.0145]
29	0.8068	0.7762	0.7838	−0.0076 [−0.0152, −0.0005]
43	0.8081	0.7881	0.7816	+0.0064 [+0.0013, +0.0113]

Source: `g2-saturation-2026-10-18/audit.json`.

4.2.3.2. Published Degformer Checkpoint and Retrained Architecture

The published Degformer checkpoint scores 0.9248, but this is not a held-out result: 42 163 of the 47 847 mutant cells (88.12%) have an exact sequence in its training file, and the wild type of all 91 peptides is in it as well. A natural suspicion is that this exact overlap alone explains the high score. Restricting the evaluation to the 5 684 cells without overlap leaves the score essentially unchanged (0.9227, Table 4.2), so exact cell overlap does not explain the gap. The remaining leak is at the level of the peptide: the model has seen most of the matrix of every peptide, and removing a few dozen cells does not create an unseen peptide. The comparison of the published checkpoint with the models of this thesis is therefore not a fair test of generalisation, and no claim is made about it.

A fair comparison of the architecture requires the same label budget and the same cluster-wise split. Retrained under these conditions (one seed, hyper-parameters as published by its authors, target the absolute PSI), the Degformer architecture reaches 0.7500, below ridge (difference −0.0582, [−0.0734, −0.0426]) and below ΔPAP (Table 4.5). This lower score is not necessarily a property of

the architecture itself: it may reflect the small number of training clusters (see Section 4.2.3.4).

4.2.3.3. Interaction Designs

Three interaction designs were fixed before running: FiLM, a rank-8 bilinear form and ridge with 2,400 cross-product features. A candidate was to be called a useful architectural component only if the following held at the same time: (a) its paired difference to ridge is positive in all three seeds, (b) the confidence interval of the ensemble difference excludes zero, and (c) the same holds against ΔPAP. None of the candidates met the criterion (Table 4.5). The best of them, FiLM, is −0.0054 ([−0.0136, +0.0028]) relative to ridge and is below ridge in every seed. The sigmoid gate of the first experiment gives the same direction, and the retrained Degformer architecture is far lower. All candidates are above the frozen ΔPAP, but so is plain ridge.

Table 4.5. Interaction designs against ridge M3. "Criterion met" applies the pre-registered rule above; for the sigmoid gate and the one-seed Degformer architecture the rule is applied descriptively.

Candidate	AUROC	Candidate − ridge [95% CI]	Per-seed difference	Criterion met
Sigmoid gate +ΔPAP (G2 runner)	0.8002	−0.0080 [−0.0152, −0.0006]	−0.0168, −0.0306, −0.0200	No
FiLM (pair-generated scale and shift)	0.8028	−0.0054 [−0.0136, +0.0028]	−0.0104, −0.0162, −0.0091	No
Rank-8 bilinear	0.7933	−0.0149 [−0.0227, −0.0072]	−0.0212, −0.0275, −0.0335	No
Ridge with 2,400 cross-product features	0.7924	−0.0158 [−0.0227, −0.0086]	deterministic	No
Degformer architecture, retrained (one seed)	0.7500	−0.0582 [−0.0734, −0.0426]	one seed only	No

Source: `g2-saturation-2026-10-18/results.json`, `g2-saturation-2026-10-18/audit.json`, `g2-extension-2026-10-18/e2-candidates.json`, `g2-extension-2026-10-18/e3-degformer-arch.json`.

This result does not show that interaction between the substituted amino acid and its context is absent. It shows that, at this sample size (89 clusters), what four different mechanisms add is not enough to offset the variance they bring. The question

of model capacity is therefore left open and is not answered by trying one more variant (see Section 4.2.3.4).

4.2.3.4. Learning Curve and Relaxing the Degron Threshold

Two questions were put together and their criteria fixed before running. **P1:** does the AUROC still rise with the number of training clusters? **P0:** can the 85 peptides removed by the degron threshold be used as extra *training* labels if the threshold is relaxed from 3.0 to 4.0? The design keeps the outer-test cells exactly equal to those of the 91 benchmark peptides, so every number is comparable with 0.8082; the added peptides enter only the outer-training sets, and those that fall in a test cluster are removed from the training of that fold (531 cells in fold 0 and 532 cells in fold 3, none in the other folds). The partition of the five folds was checked to be identical to the earlier experiment; at the 100% level, the two models reproduce the earlier scores of 0.8082 and 0.8015 exactly. Two models of different capacity were fixed in advance: ridge M3 and the gated network without ΔPAP.

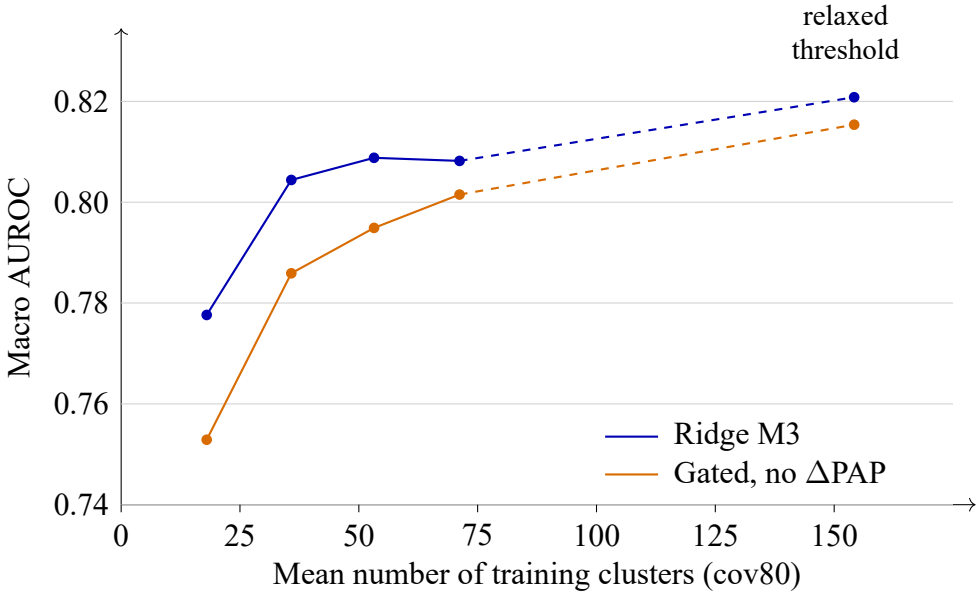


Figure 4.1. Macro AUROC against the mean number of training clusters (nested cluster subsets of 25, 50, 75 and 100% of the outer-training data; the dashed segment is the relaxed-threshold training set). Data: `learning-curve-2026-10-22/results.json`.

Table 4.6. Learning curve (P1) and relaxed degron threshold 4.0 (P0). Ranges are over 5 folds and 3 seeds.

Training level	Train clusters	Train cells	Ridge M3	Gated, no ΔPAP
25%	18	9 068–10 314	0.7776	0.7529
50%	35–36	18 635–19 877	0.8044	0.7859
75%	52–54	28 145–29 048	0.8088	0.7949
100%	70–72	38 233–38 290	0.8082	0.8015
100% + relaxed threshold	153–155	82 191–82 780	0.8208	0.8154

Source: `learning-curve-2026-10-22/results.json`.

P1. Table 4.6 and Figure 4.1 show the curves. Under the fixed rule the learning curve of ridge does not support the hypothesis of label shortage (Table 4.7). From 100% to 50% of the clusters the difference is only +0.0038 ([−0.0009, +0.0087]) and the sequence of means is not monotone. This must not be read as evidence that the data limit has been reached: the design does not measure a ceiling. The gated network behaves differently: it gains +0.0156 ([+0.0060, +0.0256]), the curve is monotone, and the criterion is met. The two curves together suggest that ridge has absorbed what the original distribution of clusters can teach it, while the larger-capacity network is still rising at about 70 clusters. The gap between ridge and the network narrows with data but is not closed (ridge is still higher by 0.0054 at the relaxed-threshold level); the result does not rehabilitate any of the rejected designs, and it must not be extrapolated to “more data will make the architecture win”. One of the predictions written before the run, that P1 would probably be supported, was therefore not borne out for ridge.

P0. Adding 85 peptides (44 490 cells, 83 clusters not in the benchmark) lifts ridge from 0.8082 to 0.8208, a gain of +0.0126 ([+0.0061, +0.0190]), positive in each seed (+0.0142, +0.0142, +0.0128). The gated network gains +0.0139 ([+0.0059, +0.0218]). Both meet the pre-registered criteria. The protocol had also predicted, before the run, that P0 would probably fail, because peptides with a high PSI_{WT} are already stable and might have almost no positive cells. The measurement contradicts this prediction (Table 4.8): the share of positive cells falls only from 13.1% to 11.4%, and the 95th percentile of ΔPSI in the added peptides is 1.51, against 1.70 in the benchmark peptides. The threshold of 3.0 therefore appears more conservative

than the mutation-effect task requires, even if it is the right threshold for the different question of whether a peptide is a degron.

Table 4.7. Pre-registered verdicts of P1 and P0

Model	Question	ΔAUROC [95% CI]	Pre-registered criteria	Verdict
Ridge M3	P1: 100% − 50%	+0.0038 [−0.0009, +0.0087]	CI> 0: no; monotone: no	insufficient evidence for label shortage
	P0: relaxed − 100%	+0.0126 [+0.0061, +0.0190]	CI> 0: yes; all seeds > 0: yes	accepted
Gated, no ΔPAP	P1: 100% − 50%	+0.0156 [+0.0060, +0.0256]	CI> 0: yes; monotone: yes	supported
	P0: relaxed − 100%	+0.0139 [+0.0059, +0.0218]	CI> 0: yes; all seeds > 0: yes	accepted

Source: `learning-curve-2026-10-22/results.json`.

Table 4.8. Cells of the benchmark peptides and of the 85 peptides added by relaxing the threshold

Statistic	Benchmark set (91 peptides)	Added set (85 peptides)
Median PSI_{WT}	2.593	3.346
Fraction of positive cells ($\Delta PSI \geq 1$)	0.1310	0.1140
Median ΔPSI	0.125	0.083
95th percentile of ΔPSI	1.697	1.507

Source: `learning-curve-2026-10-22/results.json`.

4.2.3.5. Number of Clusters or Coverage of the Distribution

The two curves of the previous section seem to contradict each other: ridge is flat between 35 and 71 clusters, yet it rises when 83 more clusters are added. One reading is that the two steps add different things: the first thickens the existing region of PSI_{WT}, the second widens it. The previous experiment cannot separate the two, because it changes the number of clusters and the region together. A second experiment, with its criteria fixed before running, keeps the whole original training set and adds a nested fraction of the new clusters (Table 4.9). The main criterion compares the gain at +50% of the new clusters (about 42 clusters) with the marginal gain of 36

old-region clusters measured before, 0.0038.

Table 4.9. Gain over the benchmark training set when a fraction of the new clusters is added

New clusters added	Mean added	Ridge	Gain	Gated	Gain
none (benchmark)	0	0.8082	—	0.8015	—
+25%	21	0.8143	+0.0061	0.8090	+0.0075
+50%	42	0.8196	+0.0114	0.8118	+0.0103
+75%	64	0.8204	+0.0122	0.8168	+0.0153
+100%	85	0.8208	+0.0126	0.8154	+0.0139

Source: `learning-curve-2026-10-22/results-b.json`.

For ridge the gain at +50% is +0.0114 ([+0.0063, +0.0166]), whose interval lies entirely above 0.0038: the main criterion supports coverage of the distribution. For the network the interval contains 0.0038, so the new clusters are useful but not shown to be better than old ones (Table 4.10). The secondary criterion, the shape of the curve, gives the opposite answer: the gain at +25% is 48.4% of the total for ridge and 54.4% for the network, below the threshold of 60% fixed for "early saturation", so the fixed rule says "proportional to the number of clusters". Because the two fixed criteria disagree, the question is *not decided*. As a descriptive remark only, the successive increments of ridge are +0.0061, +0.0053, +0.0007, +0.0005, so roughly 90% of the total gain is already reached at +50%, and the threshold of 60% at +25% was a poor operational choice. This remark does not overturn the verdict.

Table 4.10. Pre-registered verdicts on the new clusters. The reference marginal gain of old-region clusters is 0.0038.

Model	B(50%) − B(0) [95% CI]	Share of total gain at +25%	Primary criterion	Shape criterion
Ridge M3	+0.0114 [+0.0063, +0.0166]	48.4%	supports wider coverage of the distribution	proportional to cluster count
Gated, no ΔPAP	+0.0103 [+0.0033, +0.0172]	54.4%	valuable, but not shown to exceed old-region clusters	proportional to cluster count

Source: `learning-curve-2026-10-22/results-b.json`.

What can be stated is limited: clusters from the region $3.0 < \mathrm{PSI}_{\mathrm{WT}} \leq 4.0$

have a marginal value at least equal to that of old-region clusters, and for ridge the evidence suggests it is higher. It cannot be said that the gain is *only* due to wider coverage, because the 83 new clusters differ from the old ones both in region and in protein.

4.2.3.6. Sensitivity to the Threshold

The four thresholds 3.5, 4.0, 4.5 and 5.0 were run in one pass and are all reported (Table 4.11, Figure 4.2). The main configuration of the thesis stays at 4.0, a value fixed before this grid was seen and chosen then only because it roughly doubles the number of peptides; it is not re-chosen from the grid. Every threshold is better than the benchmark training set (all intervals above zero). The ridge gain is highest at 4.0 (+0.0126) and lower at 4.5 (+0.0101) and 5.0 (+0.0093); under the fixed rule (a drop of more than 0.002) this is labelled "decline after 4.0". That label describes the shape of the point estimates: the intervals overlap widely, and no interval was computed for the difference between 4.0 and 4.5. The gated network is nearly flat from 3.5 to 5.0, so the decline is a property of ridge. It also depends on how a positive cell is defined: with the positive cut at $\Delta\text{PSI} \geq 1.5$ the maximum for ridge moves to 5.0 (+0.0098) and there is no decline. The share of positive cells in the added peptides falls steadily with the threshold (Table 4.11), which is consistent with peptides that are not degrons starting to dilute the training set (an interpretation, not tested separately).

Table 4.11. Added data and gains for each degron threshold (cov80, positive cut $\Delta\text{PSI} \geq 1$)

Thr.	Peptides	Cells	New clusters	Pos. rate	Ridge AUROC	Ridge Δ [95% CI]	Gated Δ
3.5	+50	26 234	48	0.1197	0.8185	+0.0103 [+0.0043, +0.0164]	+0.0115
4.0	+85	44 490	83	0.1140	0.8208	+0.0126 [+0.0061, +0.0190]	+0.0139
4.5	+106	55 362	102	0.1028	0.8183	+0.0101 [+0.0028, +0.0172]	+0.0125
5.0	+121	63 009	117	0.0918	0.8175	+0.0093 [+0.0016, +0.0169]	+0.0133

Source: `learning-curve-2026-10-22/results-c.json`.

The gain at 4.0 survives three definitions of a positive cell and a different assignment of folds (Table 4.12); all four intervals are above zero. The last row must

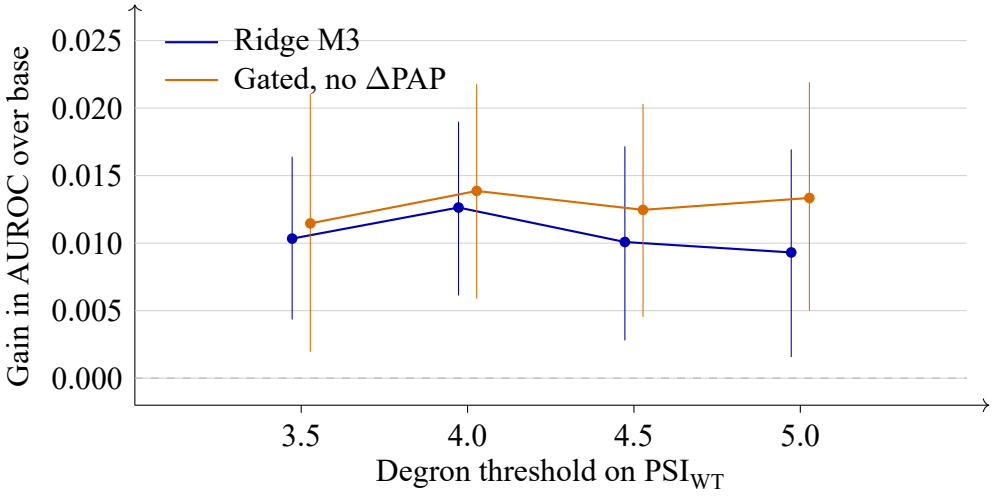


Figure 4.2. Gain in macro AUROC over the benchmark training set for each degron threshold, with 95% cluster-bootstrap intervals (horizontal offsets only separate the two models). Data: `learning-curve-2026-10-22/results-c.json`.

be read with care. The clustering at 50% coverage was meant to test sensitivity to the coarseness of clustering, but on this peptide set cov50 and cov80 produce the same number of clusters (89) and, as found after the run, the same grouping of peptides; only the representative labels differ, which changes how clusters are assigned to the five folds (the benchmark score becomes 0.8116 instead of 0.8082). That row therefore tests a re-assignment of folds, not coarser clustering; clustering granularity is tested with genuinely different partitions in Section 4.2.3.7.

Table 4.12. Robustness of the gain at threshold 4.0 (ridge M3)

Condition (threshold 4.0)	Ridge Δ vs. base [95% CI]	CI > 0
cov80, ΔPSI ≥ 0.5	+0.0110 [+0.0051, +0.0169]	yes
cov80, ΔPSI ≥ 1.0 (main)	+0.0126 [+0.0061, +0.0190]	yes
cov80, ΔPSI ≥ 1.5	+0.0094 [+0.0037, +0.0149]	yes
cov50 fold assignment, ΔPSI ≥ 1.0	+0.0096 [+0.0032, +0.0159]	yes

Source: `learning-curve-2026-10-22/results-c.json`.

4.2.3.7. Sensitivity to the Clustering Granularity

Because cov50 does not change the partition on this peptide set, a grid of seven partitions of the 176 peptides (the 91 benchmark peptides and the 85 added ones) was fixed before running: grouping by UniProt accession (no clustering), the reference MMseqs2 setting (30% identity, 80% coverage) and five looser MMseqs2 settings.

The P0 comparison (ridge M3, relaxed threshold 4.0 against the benchmark training set, same cluster bootstrap) was repeated with the folds rebuilt from each partition. The reference level reproduces the earlier scores (0.8082 and 0.8208) and the five folds of the earlier experiment. The criterion fixed in advance: the gain is "robust to granularity" if every partition that differs from the reference and is still informative (fewer than half of the added cells are lost because they share a group with a test peptide) has a bootstrap interval of the gain above zero and a positive gain in all three seeds.

Table 4.13. Gain of the relaxed threshold 4.0 (ridge M3) under different protein partitions. "Groups" are the groups of the 91 benchmark peptides and of all 176 peptides; "Shared" is the number of added peptides that fall in a group of a benchmark peptide; "Excl." is the mean percentage of the added cells that are removed from training because their group contains a test peptide.

Partition (MMseqs2)	Groups base/all	Shared	Excl. %	AUROC base → relaxed	Gain [95% CI]	Seeds > 0
Accession (no clustering)	91/176	0	0.0	0.8036 → 0.8202	+0.0166 [+0.0099, +0.0240]	yes
id 0.3, cov 0.8 (reference)	89/172	2	0.5	0.8082 → 0.8208	+0.0126 [+0.0061, +0.0190]	yes
id 0.3, cov 0.5 grouping as reference; folds assigned differently	89/172	2	0.5	0.8116 → 0.8212	+0.0096 [+0.0032, +0.0159]	yes
id 0.2, cov 0.5	87/163	6	1.4	0.8051 → 0.8185	+0.0134 [+0.0074, +0.0199]	yes
id 0.1, cov 0.5 identical to the row above	87/163	6	1.4	0.8051 → 0.8185	+0.0134 [+0.0074, +0.0199]	yes
id 0.1, cov 0.3	71/133	22	5.2	0.8096 → 0.8209	+0.0113 [+0.0037, +0.0184]	yes
id 0.0, cov 0.3 identical to the row above	71/133	22	5.2	0.8096 → 0.8209	+0.0113 [+0.0037, +0.0184]	yes

Source: `recluster-sensitivity-2026-10-22/results.json`.

The seven levels give only 4 distinct partitions (Table 4.13): lowering the identity from 0.2 to 0.1 at coverage 0.5, and from 0.1 to 0.0 at coverage 0.3, changes no

group, and 30% identity with coverage 0.5 reproduces the reference grouping (only the group labels, and hence the fold assignment, differ; the gain there is +0.0096 [+0.0032, +0.0159], the same value as the cov50 row of Table 4.12). Three partitions differ from the reference and are informative: grouping by accession (+0.0166, [+0.0099, +0.0240]), 20% identity with coverage 0.5 (+0.0134, [+0.0074, +0.0199]) and 10% identity with coverage 0.3 (+0.0113, [+0.0037, +0.0184]). All intervals are above zero and the gain is positive in all three seeds, so under the fixed rule the gain is *robust to granularity*. The point estimates are not monotone in the coarseness (the largest is for grouping by accession), the intervals overlap, and no test of their differences was planned; the absolute benchmark score also moves with the partition (0.8082 for the reference and 0.8116 for the same grouping with other folds), so absolute values must not be compared across rows.

The test is weaker than its name suggests. The coarsest partition still has 71 groups for the 91 benchmark peptides; in it, 22 added peptides share a group with a benchmark peptide and on average 5.2% of the added cells are removed from training, and one of the five test folds consists of a single group. Whether the gain would survive a still coarser split, for example by protein family, is not known.

4.2.3.8. Pre-Training on Tile Labels

The experiment of Voutsinos et al. [4] provides a degradation score for 212 616 tiles of 30 residues, without substitutions. The question was whether initialising the gated network (without ΔPAP) from a model pre-trained to predict these scores lets it learn ΔPSI better under the same label budget. The design was fixed in advance. Pre-training treats each tile as 30 wild-type rows (the substituted-residue input is zero), predicts the tile as the mean over positions and regresses the sign-reversed, standardised degron score. Two branches then share the same initial weights, the learning-rate grid (2×10^{-3} or 5×10^{-4}, chosen on the inner validation set) and the inner validation split, and differ only in whether the network starts from the pre-trained state (*pre*) or from random initialisation (*rand*). To prevent leakage, in each outer fold every gene that has a tile similar to a protein of an outer-test cluster (MMseqs2 at 30% identity with an alignment of at least 50 residues, a shared 12-mer, or the same gene name) is removed entirely from the pre-training data *before* any learning; this removed 318, 34, 176, 99, 44 genes in the five folds. The gate on the reproduction of earlier scores

was passed (ridge 0.8082 and 0.8208; the gated network within 6.0e-05 of the earlier values). Two comparisons were fixed, both with the rule "bootstrap interval above zero and a positive difference in every seed": (A) *pre* against *rand*, and (B) *pre* against ridge M3.

Table 4.14. Macro AUROC with and without pre-training on tile labels

Training configuration	Ridge M3	Network, random init.	Network, pre-trained
Relaxed threshold 4.0 (main)	0.8208	0.8175	0.8098
Threshold 3.0 (secondary)	0.8082	0.8009	0.7726

Source: `pretrain-p3-2026-10-22/results.json`.

Table 4.15. Pre-registered comparisons for pre-training. The main configuration is the relaxed threshold 4.0; the threshold 3.0 is reported with the same criteria and was not used to choose the configuration.

Thr.	Comparison	Difference [95% CI]	Per-seed difference	Met
4.0 (main)	A: pre-trained − random init.	−0.0077 [−0.0163, +0.0006]	−0.0180, −0.0259, −0.0121	no
	B: pre-trained − ridge M3	−0.0110 [−0.0197, −0.0024]	−0.0234, −0.0330, −0.0263	no
3.0 (secondary)	A: pre-trained − random init.	−0.0283 [−0.0426, −0.0149]	−0.0295, −0.0345, −0.0427	no
	B: pre-trained − ridge M3	−0.0356 [−0.0509, −0.0215]	−0.0458, −0.0550, −0.0637	no

Source: `pretrain-p3-2026-10-22/results.json`.

The result is negative (Tables 4.14 and 4.15). In the main configuration the pre-trained network reaches 0.8098, against 0.8175 for random initialisation: difference A is −0.0077 [−0.0163, +0.0006], an interval that contains zero, so pre-training cannot be distinguished from random initialisation; difference B is −0.0110 [−0.0197, −0.0024], so the pre-trained network is below ridge. At threshold 3.0 both differences are clearly negative (A: −0.0283 [−0.0426, −0.0149]; B: −0.0356 [−0.0509, −0.0215]). The difference is negative in all three seeds in every comparison. By the mapping fixed in advance, the conclusion is "insufficient evidence for a benefit of pre-training". The pre-training task itself was learned (Pearson correlation 0.864 between prediction and score on held-out genes, range 0.857–0.868 over the fifteen runs), so the failure is in the transfer to ΔPSI and not in learning the tile scores. This result

rejects only this design, in which the pre-training sees wild-type sequences only and the whole network is then fine-tuned; designs that pre-train with substitutions, freeze part of the network or train both tasks jointly were not tried, and nothing is inferred about them.

4.2.3.9. Calibration of the Scores and Ranking Utility

The score of ridge M3 is a predicted ΔPSI, not a probability. To report the probability that a substitution is positive ($\Delta\text{PSI} \geq 1$) the model trained at the relaxed threshold 4.0 was kept fixed and evaluated on the same 47 847 outer-test cells (91 peptides, 89 clusters). Calibrators were fitted only on out-of-fold scores of the outer-training pool: the pool was split again into five inner folds by cluster, the ridge model was refitted on four of them, and the scores of the fifth were kept; the outer-test cells never entered a calibrator. Three conversions were fixed in advance: the *raw* score clipped to $[0, 1]$, Platt scaling (a logistic regression on the score) and isotonic regression, with a constant forecast of the base rate of the training pool as a reference for skill. The criteria were also fixed in advance: a conversion is *calibrated* if the bootstrap interval of $\text{Brier}(\text{raw}) - \text{Brier}(c)$ is above zero and the difference is positive in every seed; it has *skill over the base rate* if the interval of $\text{Brier}(\text{constant}) - \text{Brier}(c)$ is above zero; Platt is used in the prototype unless isotonic regression is better with an interval above zero; and the ranking is *enriching* if the interval of the enrichment at 10% is above 1. The raw scores of the three seeds reproduce the earlier AUROC values (gate passed).

Table 4.16. Probability quality on the outer-test cells (observed fraction of positive cells: 0.131). ECE: expected calibration error over ten equal-mass bins; $\hat{p}$: predicted probability.

Score to probability	Brier	95% CI	Skill vs. base rate	ECE	Mean $\hat{p}$
Raw score, clipped to $[0, 1]$	0.1272	[0.1186, 0.1360]	−0.116	0.1437	0.2617
Platt (logistic) scaling	0.0925	[0.0809, 0.1046]	+0.188	0.0163	0.1146
Isotonic regression	0.0925	[0.0810, 0.1045]	+0.188	0.0164	0.1145
Constant (base rate of the training pool)	0.1139	[0.1011, 0.1276]	—	0.0154	0.1219

Source: `calibration-p6-2026-10-22/results.json`.

Table 4.17. Pre-registered comparisons of the Brier score

Comparison (positive favours the second term)	Difference [95% CI]	Per-seed difference
Brier(raw) − Brier(Platt)	+0.0346 [+0.0266, +0.0425]	+0.0346, +0.0347, +0.0347
Brier(raw) − Brier(isotonic)	+0.0346 [+0.0267, +0.0424]	+0.0345, +0.0347, +0.0347
Brier(constant) − Brier(Platt)	+0.0214 [+0.0175, +0.0253]	+0.0212, +0.0215, +0.0215
Brier(Platt) − Brier(isotonic)	+0.0000 [−0.0002, +0.0002]	—

Source: `calibration-p6-2026-10-22/results.json`.

Platt scaling meets the criterion (Tables 4.16 and 4.17): the Brier score falls from 0.1272 to 0.0925, a difference of +0.0346 [+0.0266, +0.0425], positive in each seed (+0.0346, +0.0347, +0.0347); isotonic regression is practically identical (Platt − isotonic: +0.00002, [−0.0002, +0.0002]), so Platt scaling, with two parameters, is used. The skill over the base rate is +0.188, with the interval of +0.0214 [+0.0175, +0.0253] above zero. The first criterion is a low bar: the raw score read as a probability is *worse* than a constant forecast (skill −0.116; mean value 0.262 against 0.131 observed; recalibration slope 0.35), because it is a regression output. The more informative statement is the skill of about +0.188. The ECE alone does not separate the methods here, since the constant forecast has an ECE (Table 4.16) close to that of Platt scaling.

Local calibration is not perfect. The Platt probabilities are lower than the observed frequency in every bin: the mean is 0.115 against 0.131, and in the highest bin the mean prediction is 0.434 against an observed 0.495 (interval 0.449–0.538). A likely cause is a shift of domain introduced by the relaxed threshold: the calibrator is fitted on a pool that includes the added peptides, in which the fraction of positive cells is lower than in the benchmark peptides (Table 4.8). A sensitivity analysis that was not used for the verdict, fitting the calibrator only on benchmark-domain cells, gives a Brier score of 0.0921, an ECE of 0.0044 and a mean probability of 0.131. For ordering substitutions the shift is irrelevant (the transform is monotone); for absolute probabilities it must be kept in mind.

For prioritising substitutions (Table 4.18), choosing the top 10% of each peptide (about 54 substitutions) gives a precision of 0.47 and a recall of 0.41, an en-

Table 4.18. Ranking utility: the substitutions of each peptide (532 per 28-residue peptide) are ordered by the raw score. A random ranking has enrichment 1.

Top fraction	Cells	Precision	Recall	Enrichment	95% CI
5%	27	0.551	0.241	4.75	[4.23, 5.28]
10%	54	0.466	0.407	4.02	[3.70, 4.35]
20%	107	0.371	0.611	3.04	[2.86, 3.21]

Source: `calibration-p6-2026-10-22/results.json`.

richment of 4.02 ([3.70, 4.35]); the criterion is met. The only reference is a random ranking: it is not known how much better the model is than a simple rule, for example one based on the hydrophobicity of the substituted residue. Moreover, the final model is fitted on all of the pool whereas the calibrator is learned from scores of models fitted on four fifths of it (a known limitation of cross-fitting), and all of this is still a development benchmark. The word "calibrated" here means that the Brier score is better than that of the raw score with an interval above zero; it does not mean perfect calibration.

4.2.3.10. External GPS Labels: Koren et al. Table S4

The saturation-mutagenesis tables of Koren et al. [2] use the same endpoint (PSI) and were audited as a possible source of additional training labels. The supplementary tables were downloaded manually, because the automatic download is blocked by the host. The article is not in the open-access subset of the repository, so the tables are used internally only and are not redistributed here. The gate fixed in advance required all of four conditions (Table 4.19).

The data themselves are sound: PSI is reproduced from the bin counts in all rows, an internal wild-type reference can be built for all substrates, and the direction of the effect of the last glycine agrees with the biology. The gate nevertheless fails on the number of new clusters. Table S4 holds 1 739 substitution cells for 44 substrates; the peptides are 23-residue C-terminal sequences with a stop codon in which only the last ten positions are mutated, a library different from the 28-residue internal peptides of the main labels; PSI is computed from 3 bins and not 6, so ΔPSI values from the two sources cannot simply be added; only 3.0% of the cells have $\Delta\text{PSI} \geq 1$ (against 13.1% in the benchmark), and the effects sit in the last one to three positions. Only 6 of the 13 sequences could be mapped to Swiss-Prot, and one of them overlaps

Table 4.19. Gate fixed in advance for merging the labels of Koren et al. (Table S4) into training

Criterion	Pre-registered threshold	Observed	Met
G1: new clusters at cov80	≥ 20	5 (6 of 13 sequences mapped)	No
R1: PSI recomputed from bin counts	$\lvert\Delta\rvert \leq$ 1e-06 in $\geq 99\%$ of rows	1827 of 1827 rows (max. 5.0e-12)	Yes
R2: internal wild-type reference	≥ 8 synonymous cells in ≥ 7 substrates	9 of 9 substrates	Yes
R3: direction of ΔPSI (last Gly of Clone#8)	median $\geq +0.5$, $\geq 90\%$ positive	median +1.04, 100% positive ($n = 19$)	Yes
Gate (G1 and R1 and R2 and R3)			**No**

Source: `koren-s4-audit-2026-10-22/results.json`.

with the training peptides, which leaves 5 new clusters, against the 20 required. As a diagnostic, not part of the gate, the agreement between Tables S3 and S4 on the 12 cells they share is low (Spearman 0.10), so S3 is not used as a label source.

By the consequence fixed in advance, these labels are *not* merged into training, and the planned merging of external labels (P4) and the repeated learning curve on the enlarged set (P5) were therefore not run. The table remains a possible external test set for a C-terminal degron class; this was not done in the thesis and no number is reported for it.

4.2.3.11. Scope Comparison with MetaDegron

MetaDegron [8] was audited as a possible competitor and label source. Its labels are binary motif-level annotations of curated degrons, not ΔPSI; the audit therefore compares scope and data, not predictive scores on the GPS matrix, because MetaDegron-D could not be run on GPS peptides (Table 4.20). Among the main findings, the curated positive set has only 227 human rows, and almost no region of it coincides with the GPS peptides used here (2 scanning peptides, 1 saturation peptide and 0 from the CRISPR screens). Only 1 E3 motif group keeps at least five samples after the strictest filter, so “21 E3 ligases” is a list of motif-compatible assignments, not 21 independently labelled classes. The reported MetaDegron-X AUC of 0.9670 was recomputed from the stored trees, but only on the authors’ prepared tables, and 8 of the 20 positive test coordinates are in the property table of the fitting data; it is not

an independent result and not a GPS result. A simple composition logistic baseline on the same positives, split by protein cluster at cov80, gives AUROC between 0.728 and 0.731 (core composition) and between 0.737 and 0.747 (29-residue context), on a reconstructed population and not on the authors' own training set. The audit therefore suggests positioning MetaDegron as a complementary tool for motif and E3 information rather than a head-to-head competitor on residue-level effect prediction; binary motif labels are not converted into ΔPSI.

Table 4.20. Audit of the MetaDegron data and overlap with the data of this thesis

Audit item	Value
Curated positive rows / distinct accessions	302 / 240
Rows with species suffix HUMAN	227
Rows in the property table used by MetaDegron-X	212
E3 motif groups with ≥ 5 samples: before filtering / after the strictest filter (human only, no regional or cluster overlap)	13 / 1
Same-region overlap with GPS scanning peptides (2 624 peptides)	2
Same-region overlap with saturation peptides (248 mapped)	1
Same-region overlap with CRISPR-screen peptides (31 peptides)	0
MetaDegron-X AUC recomputed from the stored trees on the authors' prepared tables (not a GPS result)	0.9670
Positive test coordinates also present in the property table of the fitting data	8 / 20

Source: `metadegron-analysis-2026-10-21/dataset-audit.json`, `metadegron-analysis-2026-10-21/overlap-audit.json`, `metadegron-analysis-2026-10-21/safe-x-reproduction.json`.

4.2.3.12. Experiments Not Done

The following parts of the plan have no results in this version of the thesis; no number is given for them.

- **Merging external GPS labels and repeating the learning curve on the enlarged set.** Not run: the only audited source (Table 4.19) did not pass its gate. Other candidate sources (for example Timms et al., Shimshon et al., Bekturova et al. and a recent preprint on disease mutations that disrupt degrons) have not been audited, and their terms of use have not been read.
- **Independent validation.** All numbers are out-of-fold on a development benchmark. Testing on labels that did not take part in any design decision, including

the C-terminal data of Table 4.19, has not been done. **[PENDING]**

- **Other pre-training designs** (with substitutions, partial freezing, joint training): not tried (Section 4.2.3.8).
- **Comparison of the ranking with simple rules.** Only a random ranking was used as reference (Section 4.2.3.9).
- **Separating the region of PSI_{WT} from the protein.** It needs pairs of peptides of the same protein on both sides of the threshold, which the present data may not contain (Section 4.2.3.5).
- **Graphical prototype and its tests, and tests with users.** Not built (Section 4.3). **[PENDING]**

4.2.4. Discussion

What the measurements support. (i) A ridge regression on 180 features is a strong baseline for the substitution effect: it is above the frozen ΔPAP baseline with an interval that excludes zero, and none of the five larger designs tried (the sigmoid-gated networks, FiLM, the bilinear form, the cross-product ridge and the retrained transformer architecture) beat it under the fixed criteria. (ii) Training labels from peptides just above the degron threshold improve the out-of-fold AUROC on the original test cells, and this gain is not caused by an obvious collapse of the label. (iii) The gain is not simply proportional to the number of clusters, but the data do not yet separate wider coverage from more clusters. (iv) The gain from the relaxed threshold does not depend on how proteins are grouped: it is above zero under every informative partition tried, including grouping by accession. (v) The score of ridge M3 can be turned into a probability with a Platt transform that is better than the raw score and than the base rate, and the top 10% of the substitutions of a peptide contains about four times more positive cells than a random choice.

What they do not support. The thesis does not claim a new architecture: no architectural component earned a place under the fixed criteria, and the gate even changes sign between seeds. It does not claim that the published Degformer is worse than the models here, nor better: its checkpoint is not a held-out result, and the retrained

architecture was run for one seed. It does not claim that more labels make a larger model win; the gated network is still rising, but it is below ridge at every level that was run. It does not claim that pre-training on tile labels helps: the one design tried was not better than random initialisation and was below ridge. It does not claim that the external labels of Koren et al. can be merged: they did not pass the gate. It makes no claim of external validation of the probabilities or of the ranking.

Threats to validity.

- The benchmark is a development benchmark on a data set that has been examined several times; the out-of-fold numbers are not an external validation, and the best figure (0.8208) must not be quoted as one.
- The three seeds share one outer partition, and the confidence intervals are conditional on the stored predictions. No multiple-comparison correction is applied.
- The threshold of 4.0 was fixed before the sensitivity grid, but only after the benchmark and a first look at the number of peptides it adds; the near coincidence with the maximum of the grid is not evidence that the way it was chosen was sound.
- All labels come from one assay and one library. Whether the gain transfers to other degron classes or to other assays is untested.
- The added peptides pass through the same label pipeline and so inherit its assumptions (the $\mathrm{PSI}_{\mathrm{WT}}$ reference from synonymous cells, the average of two halves).
- The coarsest partition tested still has 71 groups for 91 peptides, so robustness to a split by protein family is untested; the absolute scores change with the partition.
- The Platt calibrator is fitted on a pool that includes the added peptides, and its probabilities are about 13% too low for the benchmark peptides; the criterion “better than the raw score” is a low bar, because the raw score is worse than the base rate.

- The pre-training result concerns one design and one encoder; the ranking is compared only with a random ranking.

4.2.5. Provenance of the Reported Numbers

All numbers in this section are produced by `assets/ch4/build_ch4_assets.py`, which reads the JSON files below (paths relative to `topics/degron-footprint/research/`) and writes the tables, figures and number macros used here. The tables repeat their source in a note.

Table 4.21. Source files of the tables and figures of Section 4.2

Table / figure	Result files
Table 4.1	`next-steps-plan-2026-10-22/sat-filter-dropoff.json`
Tables 4.2–4.4	`g2-saturation-2026-10-18/results.json`, `g2-saturation-2026-10-18/audit.json`, `g2-extension-2026-10-18/e1-overlap.json`
Table 4.5	`g2-extension-2026-10-18/e2-candidates.json`, `g2-extension-2026-10-18/e3-degformer-arch.json`
Figure 4.1, Tables 4.6–4.8	`learning-curve-2026-10-22/results.json`
Tables 4.9, 4.10	`learning-curve-2026-10-22/results-b.json`
Figure 4.2, Tables 4.11, 4.12	`learning-curve-2026-10-22/results-c.json`
Table 4.20	`metadegron-analysis-2026-10-21/dataset-audit.json`, `metadegron-analysis-2026-10-21/overlap-audit.json`, `metadegron-analysis-2026-10-21/safe-x-reproduction.json`, `metadegron-analysis-2026-10-21/composition-baseline.json`
Tables 4.13	`recluster-sensitivity-2026-10-22/results.json`
Tables 4.14, 4.15	`pretrain-p3-2026-10-22/results.json`
Tables 4.16, 4.17, 4.18	`calibration-p6-2026-10-22/results.json`
Table 4.19	`koren-s4-audit-2026-10-22/results.json`
Table 4.22	`calibration-p6-2026-10-22/verify.json`
Measurement ceiling (Section 4.2)	`residue-task-derisk-2026-10-15/ceiling.json`

4.3. Software Testing Plan

4.3.1. Functions Under Test

UC-01 to UC-03 of Chapter 1. The only implemented prototype is a minimal offline command-line tool (`predict.py` in `research/calibration-p6-2026-10-22/`) built on the model of Section 4.2.3.9. It covers the functions as follows. UC-01 (effect matrix): it scores all 532 substitutions (28 positions $\times$ 19 amino acids) of a 28-residue tile, given as a sequence or as a tile of a UniProt accession from the local snapshot. UC-02 (ranking): the output gives the rank of each substitution within the tile and a flag for the top 10%. UC-03 (export): the output is written as CSV with the raw score and the calibrated probability. Not implemented: JSON export, the model version and data source as fields of the exported file (a warning with the supported domain is printed to the error stream only), the graphical interface, and a block on tiles outside the supported domain.

4.3.2. Approach

The prototype is a thin layer over a research model, so the main risk is that it computes something different from what was evaluated in Section 4.2. The approach is therefore *back-to-back testing*: the prototype path is compared with the training and evaluation path of the research runner on the 92 337 cells of the 176 peptides. The 180 features are rebuilt in the prototype by an independent implementation (16 620 660 values are compared), so the comparison is not a code path checked against itself. Black-box tests of input validation (wrong length, characters outside the 20 amino acids, unknown accession, tile outside the protein) are planned but have not yet been written as automated tests **[PENDING]**.

4.3.3. Criteria for Test Success or Failure

A check passes when the largest absolute difference between the two paths is below 10^{-9}. The check of the exported file is asserted at 5×10^{-9} in the code, because values are written with ten significant digits; the measured differences are below 10^{-9} for it as well. A test fails when any difference exceeds its limit.

4.3.4. Suspension Criteria and Restart Requirements

The verification script stops at the first failed assertion and runs no later check. The command-line tool refuses to run if the model file is missing. Restart: rebuild the model file with `build_model.py` and run the verification script again from the

beginning.

4.3.5. Test Deliverables

The verification script (`verify.py`), its machine-readable report (`verify.json`), the calibration results (`results.json`), and Table 4.22 of this chapter.

4.4. Test Management

4.4.1. Test Activities

Executed: building the model and the caches, running the verification script (11 checks). Not yet executed: input-validation tests, tests of the interface, performance measurement, and tests with users **[PENDING]**.

4.4.2. Test Environment

Linux under WSL2 on one development computer, CPU only; Python 3.12.3 and NumPy 2.5.3. The verification needs about 1.2 GB of memory. The model file and the caches are generated by the scripts and are not stored in the repository.

4.4.3. Resources and Allocation

One tester, the author of the thesis, who is also the developer; there is no independent tester.

4.4.4. Risks

- Back-to-back tests show agreement with the research code, not that the science is right: an error shared by both paths (for example in the definition of the label) would not be found.
- The probabilities are about 13% too low for peptides of the original benchmark domain (Section 4.2.3.9); ranking is not affected.
- The tool does not know the wild-type PSI of a user's tile, so a stable tile outside the supported domain still receives a score; only a printed warning protects the user.
- The model was evaluated only on a development benchmark; no result in this chapter supports a claim of external validity.

4.5. Test Scenarios and Test Cases

4.5.1. Test Scenarios

S1: score a tile given as a sequence. S2: score a tile of a protein given by its UniProt accession. S3: export the scores as CSV and read them back. S4: reject invalid input (planned **[PENDING]**).

4.5.2. Test Cases

Table 4.22 lists the executed cases, which cover S1 to S3 through the comparison of the scores computed by the prototype with those of the research path. The cases of S4 (length different from 28, a character outside the 20 amino acids, an unknown accession, a tile beyond the end of the protein, a tile with a non-standard residue, a missing model file) are specified by the input checks in the tool but have no automated test yet **[PENDING]**.

Table 4.22. Executed test cases: agreement of the prototype with the research runner. The tolerance is 10^{-9}.

Test case (back-to-back consistency check)	Max. abs. difference	Passed
TC-01: Labels: label pipeline vs. cached labels	0.0e+00	Yes
TC-02: Features: independent featuriser in the prototype vs. cached matrix	1.1e-16	Yes
TC-03: Raw score: prototype vs. trained ridge model	4.4e-16	Yes
TC-04: Platt probability: prototype vs. scikit-learn	3.3e-16	Yes
TC-05: Isotonic probability: prototype vs. scikit-learn	1.7e-13	Yes
TC-06: UniProt accession → tile → score: vs. training path	0.0e+00	Yes
TC-07: UniProt accession → tile → probability: vs. training path	0.0e+00	Yes
TC-08: Exported CSV, raw score: vs. in-memory prediction	4.4e-10	Yes
TC-09: Exported CSV, probability: vs. in-memory prediction	4.8e-11	Yes
TC-10: Features of the 91 benchmark peptides: prototype vs. cache	1.1e-16	Yes
TC-11: Outer-test scores of all 15 (seed, fold) pairs: prototype path vs. runner	4.4e-16	Yes

Source: `calibration-p6-2026-10-22/verify.json`.

4.6. Test Evaluation

All 11 executed cases passed; the largest absolute difference is 4.4e-10 (Table 4.22). The prototype therefore reproduces the research model on 92 337 cells,

including the scores of the outer-test folds of all fifteen (seed, fold) pairs. This establishes consistency only. It does not show that the predictions are correct (Section 4.2 addresses that, on a development benchmark), and it says nothing about input validation, the interface, the response time (the target of Chapter 1 has not been measured) or usability **[PENDING]**.

PART 3. CONCLUSION

1. Achieved Results

1.1. Knowledge and Skills

[PENDING: to be written from the results actually achieved.]

1.2. Program and Practical Application Potential

[PENDING]

2. Limitations and Challenges

[PENDING]

3. Future Work

[PENDING]

REFERENCES

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APPENDIX A: INSTALLATION AND USER GUIDE

A.1. Installation Requirements

A.1.1. Hardware Requirements

[PENDING: measured requirements for the research runner and the prototype.]

A.1.2. Software Requirements

Python with the packages listed in the project's requirements file, a CPU build of PyTorch, and MMseqs2 for clustering. **[PENDING: exact versions.]**

A.2. System Deployment

[PENDING]

A.3. User Guide

[PENDING: after the prototype exists.]

APPENDIX B: SUPPLEMENTARY MATERIALS

B.1. Source Code and Data Availability

[PENDING: repository link and release; data are redistributed only where the terms of use allow.]

B.2. Additional Tables and Figures

[PENDING]